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Methodology to quantify crop, irrigation water, and energy linkages in small-holder irrigation systems

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Abstract

In India, balancing the conflicting objectives of reducing irrigation water and energy usage, food security, and farmer income, has become very important due to constrained water availability, especially in the context of subsidies for irrigation energy. Policies to reduce water and energy usage are being proposed and tested without sufficient data or analysis in support of the expected outcomes. Some examples of these policies are energy payback schemes or promotion of sprinkler and drip, to control water usage,

Most studies in the water-energy-food nexus (WEFN) area have been conducted using simplified or aggregate data, thus missing the nuances that will arise due to variations in factors such as type of water source, irrigation method, and extent of water usage. This work develops a methodology codifying the interlinkages between crop type, irrigation water usage, and energy usage, at the farm level for Maharashtra state, India. Thus satisfying an important demand in the nexus formulation.

The methodology has two types of parameters. One set of parameters are shown to be uniform across the state. Broadly these are: (i) energy intensity as a function of water source (ii) irrigation depth per irrigation event as a function of irrigation method. These are developed from field measurements.

A second set of parameters incorporate site-to-site variations and are developed through observation. Broadly these are: (i) number of irrigation events per season for a crop (ii) crop to water source correlation.

To apply the methodology in a specific area, the values for the second set of parameters need to be determined through survey. The outputs are energy usage in kWh/ha and water usage in mm in the area for a particular crop. In this paper, the methodology has been applied at five locations each the size of a few villages.

The methodology results illustrate broad trends and distinctions in energy intensity (kWh/m^3), energy usage (kWh/ha) and water usage (m^3/ha) categorized by monsoonal (kharif), winter (rabi), plantation crops, and sugarcane, and by groundwater and surface water sources. Examples illustrating the potential effects of proposed policies in light of the results are provided. The

crop-wise cost of energy as a percentage of profit is calculated to understand the economic value of subsidies.

The methodology reveals the complex irrigation practices and factors that determine energy usage in irrigation. Its generalized nature ensures that it can be applied across the state and is not limited to site-specific cases. The work highlights the importance of conducting research and developing datasets, at a local level.

Key Words:
Water - energy - food nexus; Water-energy productivity; Local irrigation practices; Efficiency of pumps; Drip and sprinkler irrigation; India

1. Introduction

In India agriculture is a livelihood for 55% of the population, responsible for 70% of freshwater withdrawals, and 18% of the total electricity usage. Subsidized electricity together with other agricultural policies have led to groundwater depletion, seasonal water scarcity, poor quality of electricity supply to farmers, and deleterious effect on the financial health of electricity utilities (Mukherji, 2022; Barik et al., 2017; Kimmich, 2013; Dubash, 2007). Various policies are being promoted to optimize water usage in agriculture so as to balance the conflicting objectives of farmer livelihood, food security, and sustainable and efficient use of water and energy. However there is insufficient data and analysis to support the expected outcomes of the policies. Studies have proposed or analyzed interventions such as reducing water usage through crop choice (Mukherji, 2022), increasing irrigation efficiency (Fishman et al., 2015), suitable energy pricing (Kumar, 2005; Kumar et al., 2013; Mukherji, 2022), pay back schemes for reduced energy usage (EA, 2020; Chaba, 2019; Nair, 2016), and energy supply management (Shah et al., 2008). These interventions have been analyzed with secondary data at the scale of a district or larger, in general terms without quantification, or without consideration of hydrogeological conditions, and without field level measurements.

Researchers as well as policy makers have pointed out the need to consider regional variations in hydrogeology, and agro-climatic conditions (Fishman et al., 2016; Pachauri and Mehrotra, 2002). Dholakia (2020), has called for collection of data on crop-wise inputs including electricity at the state or district level for the subnational accounts, so as to provide a firm foundation for policy development. The importance of collecting site-specific, empirical data to address local variations has been noted by water-energy-food nexus (WEFN) researchers (Zhang et al., 2018a; Stillwell et al., 2011).

The scientific need for localization is also clear. Crop choices, water sources, irrigation practices, and the resultant energy use, are linked to each other in ways that have variations at the local level

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which need considerations beyond the usual parameters such as groundwater depth, crop evapotranspiration, and irrigation method. For example, irrigation has evolved from a simple pumping from a well to the field, into complicated water transfers (Kumar et al., 2022; Joshi, 2017; Kale, 2017). Hence using groundwater depth to gauge energy usage as implemented by Chen et al. (2019) at a county level (as in the North China plain) is not sufficient here. Dryland crops comprise a large proportion in India and there is a lot of variation in the water applied to these crops. A crop may be rainfed and cultivated in deficit conditions, or irrigated to full requirement, depending on a farmer's access to water (Singh et al., 2020; Bhopale, 2019; Reindl et al., 2019; Nimbalkar et al., 2016;). Hence potential evapotranspiration is not sufficient to gauge irrigation water applied to dryland crops.

This work addresses this gap by codifying the interlinkages between crops, energy usage, and water usage, through a set of parameters that are a rational reflection of the hydrogeology, agro-climatology, irrigation technology, and local irrigation practices, in Maharashtra state. The parameters are developed based on field level energy and water usage measurements and a detailed investigation of irrigation practices. These parameters are used to find the crop-wise energy and water usage. The need for such an approach is illustrated through an example below.

Common methods to estimate energy intensity, or energy used per unit of irrigation water applied (kWh/m^3), in WEFN studies involve a single pump efficiency value and water table level in the region, or a calculation from water and energy usage data aggregated at the district or country level (Campana et al., 2022; Zuo et al., 2021; Zhang et al., 2018b; Miralles-Wilhelm, 2016).

However, this work reveals that the energy intensity for wheat, rabi onion and orange, is as high as 1.4 kWh/m^3, whereas for gram and cotton it is between 0.2 and 0.5 kWh/m^3. Hence the increase in cost of electricity for cultivation of a specific crop, if energy tariffs are increased, will not be proportional to the quantity of water being used. This is important in understanding the potential effects of tariff changes or energy payback schemes to reduce water usage.

The use of pressurized irrigation increases the overall energy usage in kWh/ha by 13% in the case of gram, and reduces it by 2.5 times in the case of orange cultivation. Hence if policymakers hope to promote the use of water saving devices through energy payback schemes, there needs to be an understanding of the crop-wise variations in the effect of energy-water trade-off inherent in pressurized irrigation methods (Espinosa-Tasón et al., 2020). As Zhang et al. (2018a) note, simplifications may not capture reality thus potentially misrepresenting the final results.

The methodology used in this study may be applied to a village, district, or any area, where the parameters are applicable. For example, open well depths vary from 9 to 15 m in the state and farmers use 5 or 3 HP pumps. This configuration combined with the phenomena described above, result in average energy intensities with a certain variance, for pressurized and unpressurized irrigation methods, which are applicable across the state. In this manner a generalized methodology takes this work beyond site specific case studies.

Preprint of: https://doi.org/10.1016/j.esd.2025.101719

Site-specific variations are taken into account as illustrated in the following example. In Maharashtra borewell water is often used for perennial crops such as orange or grape since open wells begin to dry out some months after the monsoons. The total energy used for a crop will be strongly affected by the month in which borewell usage begins due to their greater depth. In a particular village and year, 40% of the open wells may dry out in October, increasing to 100% in May. This trend is codified into a parameter, thus allowing the quantification of a range or average, of kWh/ha, for post-monsoonal crops in the village.

Another example of such practices is the number of irrigation events applied to some field crops which may vary from none, to 5 or more, as mentioned earlier.

As has been pointed out by Bazilian et al. (2011), such tools quantifying WEF resource interaction aid policy design, and are an important demand in the nexus formulation.

2. Overview of the methodology

Four main parameters have been codified in the methodology and are listed in figure 1. Of these, the energy intensity and irrigation depth per event are functions of hydrogeology and irrigation technology, and are uniform across the state. These are physical parameters, and are determined through measurements. The other two parameters are related to local irrigation practices, with a wide range of variations seen among farmers. They are documented through surveys and observations. The four parameters are briefly described below.

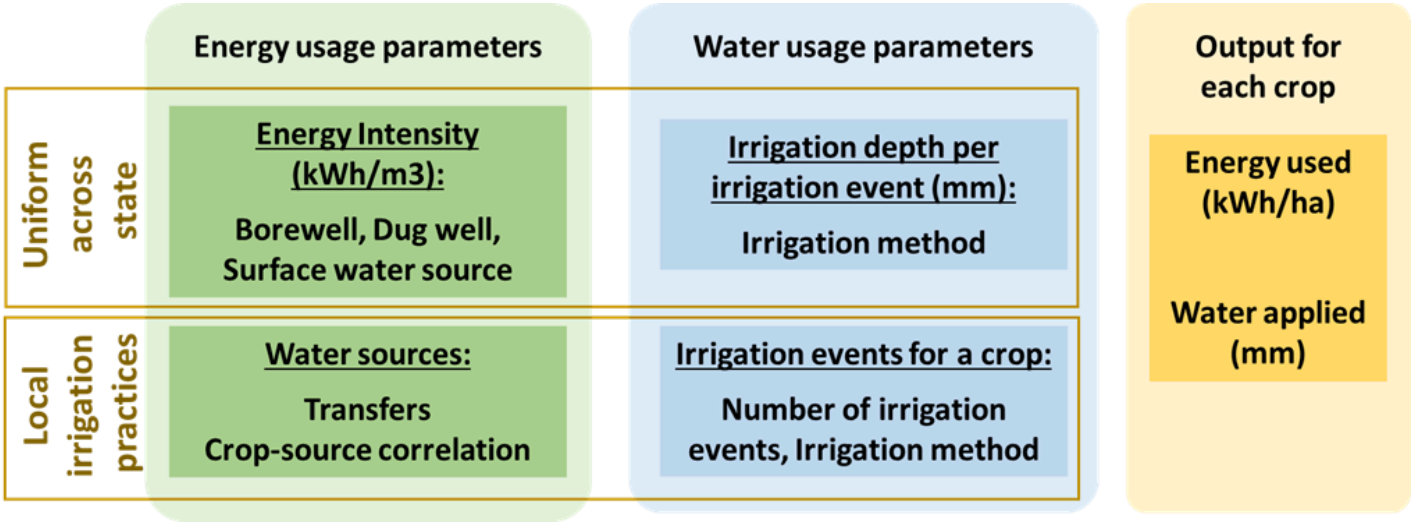


Figure 1. A block diagram listing the parameters, the main factors for each parameter, and the output of the methodology.

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Energy intensity is defined separately for each source type. Some distinctions for a water source type include pressure head as a function of irrigation method and water table change with season.

A correlation of crops to water sources is observed due to the seasonality of both crops and water sources. Water transfers across sources are carried out due to certain physical conditions. These irrigation practices can be generalized and hence are parameterized.

Irrigation depth is the height of the column of water added to the farm in one irrigation event. There may be between 1 to 30 irrigation events in a season depending on the crop, water availability, and other agro-climatic factors. The number of irrigation events is relatively easy to find through surveys, and along with irrigation depth, provides a simple way to estimate water applied.

The output of the methodology is the energy used per unit area in kWh/ha, and water applied in mm, for each crop-irrigation method combination. These water and energy usage indicators are calculated as an average at a site or village. While there is farmer to farmer variation even within these sites, the averages are representative of broad trends. For example, we find that sugarcane irrigation in the region of Western Maharashtra where sources are rivers and canals, requires 3000 to 7000 kWh / ha of energy. Whereas in Marathwada where sources are borewells, the energy usage may be estimated at 7000 to 12000 kWh / ha.

4. Study locations and methods

The map of Maharashtra state in figure 2 illustrates the study locations. Each location has between one and nine contiguous villages. In the rest of the text the locations are referred to by the cluster numbers assigned in the figure.

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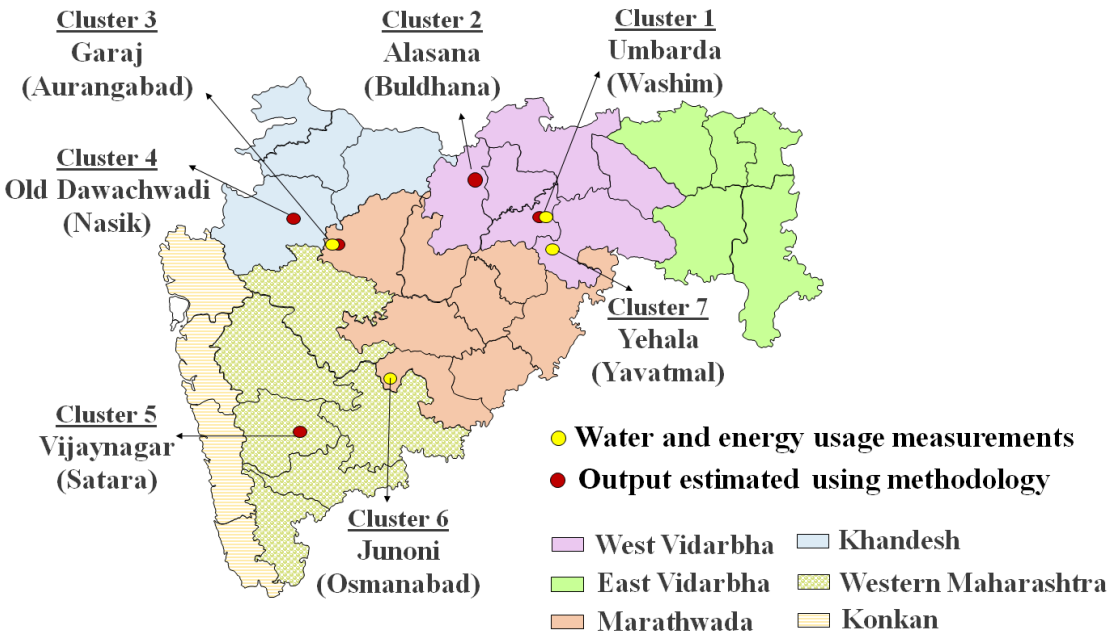


Figure 2. A map of Maharashtra. Study locations are marked as red and yellow circles. The name of each location is accompanied with the district name in parentheses. Administrative divisions are indicated by background colors. Districts are outlined by black boundaries.

Most of the data was collected from January 2020 to May 2021 as part of the 'Project on Climate Resilient Agriculture' (PoCRA) being implemented by the Department of Agriculture, Government of Maharashtra (MahaPoCRA, 2017). This is a project funded by the World Bank and carried out from 2017 to 2024 in 5000 villages of 15 districts in the divisions of Marathwada and West Vidarbha, to enhance climate-resilience and profitability of smallholder farming systems. Data in clusters 2, 4 and 5 was collected in 2016 to 2017 as part of academic projects. Main agro-climatic characteristics of the locations are listed in Table A.1 in appendix A.

There are 6 administrative divisions broadly distinguished by water sources and cropping trends (Jadhav et al 2020). Of these Konkan has almost negligible irrigated crops and East Vidarbha has less than 6% of the net irrigated area in the state. Hence our focus is on the remainder of the four divisions.

4.1 Clusters where methodology was applied

The methodology was applied in clusters 1 to 5, marked in the map as 'Output estimated using methodology'. These clusters cover the main distinctions of crops, water source, and irrigation practices in the state and broadly reflect the crops in each division as follows.

8 field crops with the highest cultivated areas cover 79% of the gross sown area in the state. Rice has a large area but is unirrigated. The rest of the crops have less than 4% coverage each. These 8 field crops are well represented in the study locations with each cluster representing most of the major crops of the division. Table A.2 lists the main crops in the state by division. Section A.1

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and table A.3 list and describe high level cropping data and water sources for the clusters.

Horticultural crops cover 1.5 Mha of area out of a net sown area of 16.7 Mha in the state and are important since they are often irrigation intensive (Department of Agriculture, 2019-20, Department of Horticulture, 2016-17). 6 of the major horticultural crops which constitute 62% of horticultural cultivated area in the state are represented in the 5 clusters.

80% of the irrigation in the state is from groundwater (Asoka et al., 2017). Clusters 1 to 3 are all wholly groundwater irrigated, with cluster 2 having no borewells. Clusters 4 and 5, also have surface water irrigation and represent two additional important distinctions of irrigation energy use as follows.

Cluster 4 is representative of surface irrigation used for horticultural crops. The cluster is in Nasik, the district with the highest horticulture production in India with canals being an important source of irrigation. The nature of the water source and horticulture cropping, results in extensive water transfers (Doval and Jadhav, 2020; Prasad and Sohoni, 2020; Kadbhane and Manekar, 2017).

Cluster 5 is representative of the Western Maharashtra division where 72% of sugarcane in the state is cultivated, and is irrigated using surface water sources. Sugarcane covers 34% of the gross cropped area in this cluster and is irrigated by river water.

4.2 Clusters where energy and water usage measurements were conducted

The four clusters marked as ‘Water and energy usage measurements’, are villages where water and energy measurements were carried out. These locations were selected from the subset of PoCRA villages where there was extensive disbursement of irrigation equipment, horticultural cropping, and presence of surface water source, thus ensuring a wide swath of crops and irrigation methods for measurement. Clusters 6 and 7, which are not in the 5 clusters above, were selected to ensure observation and measurement of long distance pumping from surface water sources. Table A.4 lists the high level representation of crops, irrigation methods, water sources, and water transfers, in the measurement clusters.

Energy and water usage data was collected over a year by metering 33 farmers. Fortnightly surveys of these farmers were conducted to find the number of irrigations given to each crop. On-field measurements of volume of water applied per irrigation event, pump efficiency, power consumption, and flow rate measurements, were carried out on 50 other pumps in these locations. These measurements were used to develop the energy intensity, and irrigation depth per event, parameters.

Individual farmer surveys, Focused Group Discussions (FGD), interviews with the village

Preprint of: https://doi.org/10.1016/j.esd.2025.101719

Agricultural assistants, and on-field observations were used to learn about crops cultivated, irrigation methods, availability of water sources, their seasonality, and water transfers, in all villages of the five clusters.

5. Results: Methodology parameters

5.1 Energy intensity (kWh/m^3)

5.1.1 Energy intensity of open wells

Open wells are the commonest source of water in the state with an average depth of 12.5 m, with a standard deviation of 4.5 m (GSDA, 2014). Open wells in the study locations are 9 to 15 m in depth representing the state data.

Predominantly pumps of 3 and 5 HP rating are used for all types of water sources. Farmers use individually owned pumps and plastic pipes, pumping from privately owned wells. Surface, or unpressurized, irrigation is the most common practice overall but sprinklers are commonly used for certain crops such as Gram, and drip is used for some horticultural crops and sometimes for Cotton.

102 energy intensity measurements were obtained through on-field measurements of power consumption and flow rates, and through energy and water metering of open wells. Figure 3(b) shows the energy intensity as a function of the head. The head is the sum of static, frictional, and an additional pressure head when using sprinklers or drip. The details of each data point and the head components are provided in Appendix B. Figure 3(a) shows the operational efficiency calculated from the energy intensity and head.

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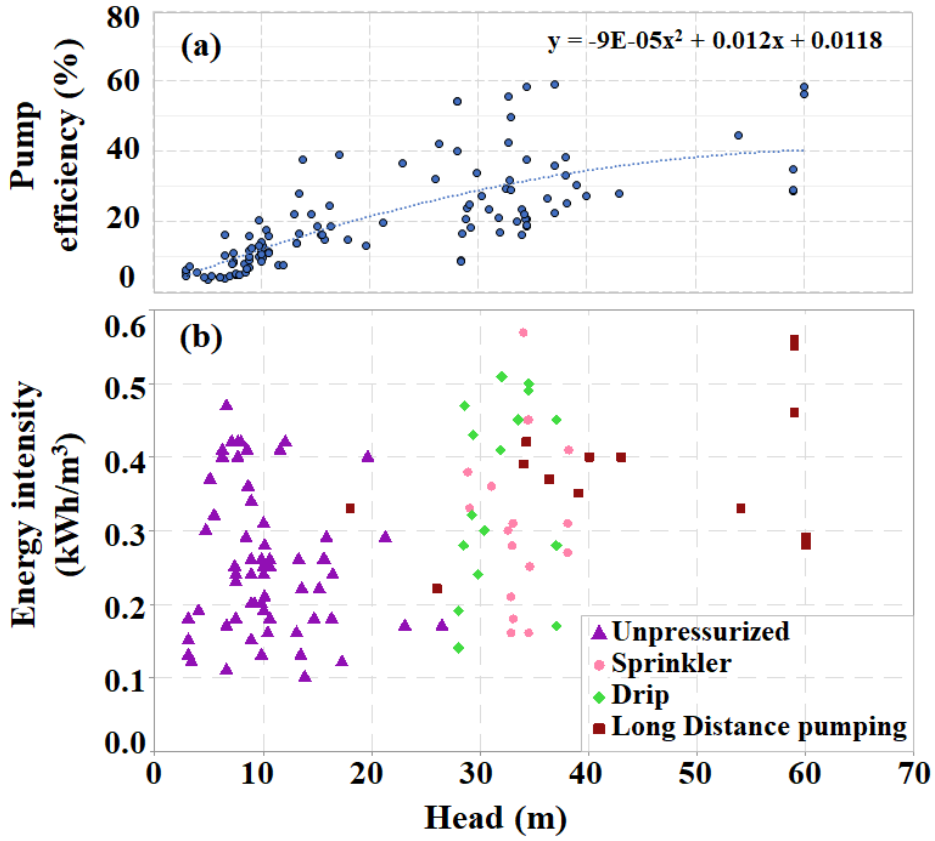


Figure 3. Operational data of pumps in open wells and surface water sources collected in the study area **(a)** Operational efficiency versus head. **(b)** Energy intensity versus head. The energy intensity of open wells is illustrated as a function of unpressurized, drip, and sprinkler irrigation. Long distance pumping refers to pumping from surface water sources, and also from open wells over a distance greater than 400 m.

The main effect observed in the efficiency data is that as the total head increases, the efficiency increases. This is because farmers purchase pumps designed to operate effectively at the maximum expected operating head, and since farmers commonly use sprinklers, the maximum expected operating head is 35 m or more since drip and sprinkler usage add a head of 25 to 30 m. For unpressurized irrigation, the operating head is a maximum of 15 m attributable to the well depth. A pump with a Best Efficiency Point at 35 m operates at a low load condition during unpressurized irrigation and hence is very inefficient. As a point of reference, the standards for agricultural pump efficiency are a maximum of 35% for 3 and 5 HP irrigation pumps at typical operating conditions (Bureau of Indian Standards, 2018).

Generally, it would be expected that the energy intensity increases directly proportional to the head. However, the efficiency effect negates this thus flattening out the relationship between head and energy intensity to some extent. Fig.3(b) shows the energy intensity data in kWh/m^3 versus

Preprint of: https://doi.org/10.1016/j.esd.2025.101719

head. For unpressurised irrigation, the data indicates a clustering between 0.1 and 0.3 kWh/ m^3, with heads between 3 and 18 m. And for pressurized irrigation there is a clustering between 0.15 to 0.45 kWh/ m^3, with heads between 28 and 40 m. Average values of 0.33 $\pm$ 0.03 kWh/ m^3 and 0.26 $\pm$ 0.02 kWh/ m^3 for a Confidence Interval of 95% were obtained for pressurized and non-pressurized irrigation respectively. These average values are used in the methodology.

Long distance pumping measurements included in the figure are discussed in section 5.1.3.

5.1.2 Energy intensity of borewells

Borewell depths in the state are largely between 60 and 110 m (MI Census, 2017) and are much less common than open wells as mentioned earlier..

In the study area borewells are 60 to 120 m in depth. The energy intensity was observed to increase from 0.5 to 1.0 kWh/ m^3 in a linear trend from November to April as shown in figure 4. Borewells are used between November and May - the reason for this is explained in section 5.4.

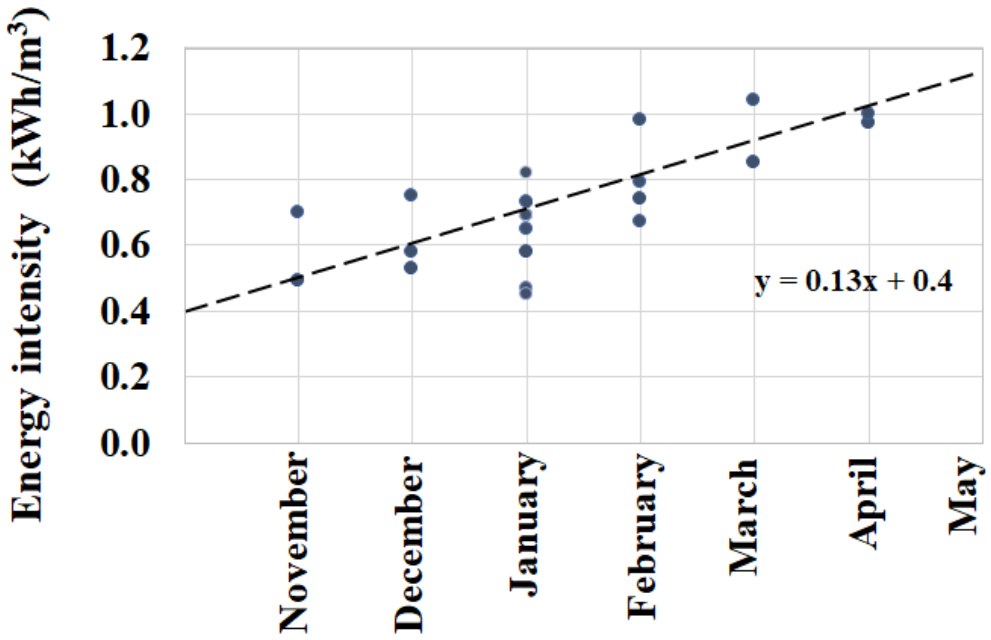


Figure 4. Energy intensity data of borewell pumping collected in the study area

The exact water level in borewells is unknown hence it is not possible to find exact efficiencies. However, they are estimated at between 27% to 32% based on expected water table levels which is commensurate with the efficiency standards for 3 and 5 HP agricultural pumps. Data for energy intensity is not available to us after April, hence the trendline it is extrapolated to May.

5.1.3 Energy intensity of surface water sources

Clusters 4, 5, 6 and 7 had surface water sources, a canal, river, dam, and irrigation tanks, respectively.

Preprint of: https://doi.org/10.1016/j.esd.2025.101719

For surface water sources, energy intensity is estimated based on head and pump efficiency, the latter considered from the trend-line in fig.3 (a). This is treated differently from open wells because the head can vary widely in case of surface water sources. The head includes the change in elevation and the frictional head which is based on typical flow rates. A satellite map for cluster 5 along with elevations is illustrated in figure 5 as an example. Details of the energy intensity estimates for clusters 4 and 5 are provided in Appendix B.

The estimated energy intensity in the study locations, range from 0.1 to 0.6 kWh/m^3 which is commensurate with measurements. Measured energy intensities range from 0.2 to 0.6 kWh/m^3 and are illustrated as 'Long distance pumping' in fig.3(b).

Elevation above sea level (m)	
1	617
2	613
3	613
4	609
5	626
6	622
7	625
8	620
9	617
10	620

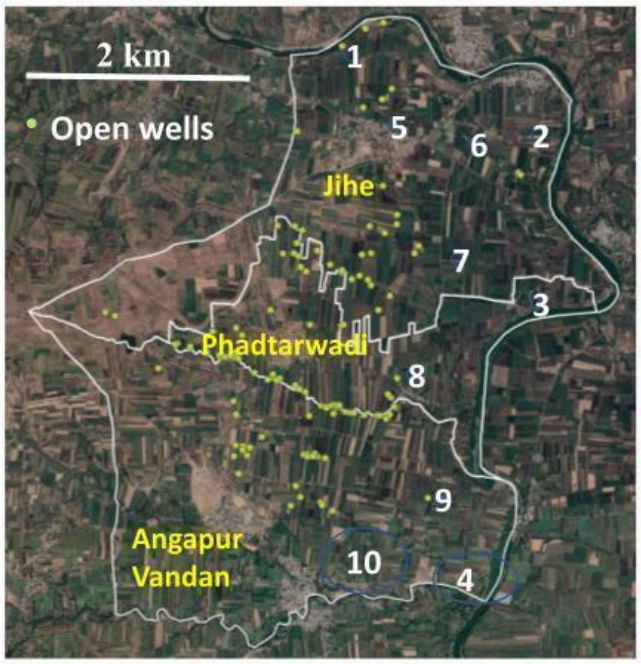


Figure 5. Satellite image of cluster 5. The river (Krishna) borders the North and East of the villages. Elevations are shown up to a distance of 1 km from the river bank. Water from the river is transported within that area.

5.2 Irrigation depth per irrigation event (mm)

Irrigation depth in an irrigation event depends on irrigation methods such as border, furrow or sprinkler, field size, dimensions and slope of the furrows and borders, soil type, and farmer practice.

In Maharashtra average land holding is 1.35 ha (Ag 2015) with 80% of farmers having less than 2 ha ownership. Irrigation is done manually, predominantly through surface methods. Common practices involve moving the pipe outlet to different locations in the field, and directing the flow by opening furrows and basins which are formed using low mud bunds. The larger fields may have manually operated chambers (hydrants) at a few locations (Khobaragade et al., 2021).

Preprint of: https://doi.org/10.1016/j.esd.2025.101719

Sprinklers are hand-move, and drip irrigation valves are manually operated. The dimensions of furrows, and borders in the study area are provided in section A.2 of Appendix A. The soil type in most of the study area is clayey, clay loam, or gravelly clay loam. The main distinctions and effects for irrigation depths observed in the study area have been listed in Table 1.

Table 1. Irrigation depths categorized as a function of irrigation method, and other conditions

Irrigation method and other conditions	Crops	Average Irrigation depth (mm)	Standard Deviation	Sample size
Border	Wheat, Gram, Maize, Sorghum, Cotton, Rabi Onion	54	13	40
Furrow	Pigeon pea, Turmeric, Sugarcane, Vegetables, Orange [a]	45	13	12
Sprinkler	Gram, Groundnut, Sesame, Soybean, Wheat	48	12	23
Sprinkler for first two irrigations of the season	Wheat, Gram	66	2.4	42
Drip with plantation crops and cotton	Orange, Banana, Grape, Cotton	17	5	8
Drip with closely spaced seasonal crops	Ginger, Onion	37	15	9

[a] For orange, first two irrigations of the season, occurring in November, are furrow, even for farmers who use drip subsequently

As a point of comparison, irrigation depth for basin, furrow, and border methods have been documented in a broad range between 30 to 70 mm for small scale irrigation (C. Brouwer et al., n.d., p. 7). More accurate data applicable to the study region could not be found in literature.

For plantation crops the number of irrigation events range from 100 to 200 over the season. In this case, the number of irrigation events cannot be used as a parameter. Hence the water applied was either measured, or if measurements were not available theoretical irrigation demand was assumed to be the water applied with the consideration of a factor for irrigation efficiency. This is a reasonable assumption borne out by observations and literature (Prasad and Sohoni, 2020). This is in contrast to most field crops which are commonly cultivated in a deficit mode in Maharashtra due to water paucity (Singh et al., 2020; Bhopale, 2019; Reindl et al., 2019; Nimbalkar et al., 2016;).

The method and data used to calculate theoretical irrigation demand are provided in Appendix B.

Preprint of: https://doi.org/10.1016/j.esd.2025.101719

5.3 Number of irrigation events and irrigation method

Tables A.5 to A.9 in Appendix A list the average number of irrigation events for each crop-irrigation method combination in the five clusters.

5.4 Water source correlation to crops

The correlation of crops to water sources is first and foremost the result of the seasonality of water sources and crops. Main crops cultivated in the state and their seasonality are listed in figure 6. The annual agricultural cycle begins with the monsoons in June. Monsoonal crops known as kharif, and long kharif field crops, cover 14.4 Mha, winter field crops known as rabi crops, cover 5.5 Mha, sugarcane 0.8, and fruit and vegetables 1.5 Mha (Department of Agriculture, 2019-20, Department of Horticulture, 2016-17). Other than wheat and sugarcane, and rice, field crops are commonly grown in a deficit irrigation mode.

	Jun	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Kharif	**Soybean, Kharif Maize, Rice Mung Bean, Black Gram,**											
Long Kharif	**Pigeon Pea, Cotton, Tomato, Green Chillies, Ginger**											
Rabi						**Wheat, Gram, Rabi Sorghum, Rabi Onion**						
12 to 18 month crops	**Sugarcane**											
Plantation crops	**Mango, Grape, Citrus, Banana**											

Figure 6. Cropping cycles for the main crops in Maharashtra. The blue squares indicate the monsoon months.

Water sources are replenished every monsoon season from July to August (Barik et. al, 2017), resulting in the water table being at its highest then, and thereafter dropping. Open wells begin to dry up from November onwards (GSDA, 2014), while borewells have water until later. This is a manifestation of shallow hard-rock aquifers that cover the state.

Hence kharif and long kharif crops are irrigated from open wells. Rabi and yearlong crops are irrigated from open wells, and when open wells dry up, from borewells or long distance pumping from surface water sources. The open well to borewell ratio is 6:1 (MI Census, 2017), in keeping with the high proportion of kharif crops compared to rabi and yearlong crops. The prioritization of open wells over borewells in kharif even by farmers who own both is explained in section 5.5 on water transfers.

An exception is Gram and rabi Sorghum, which are rabi crops, yet are irrigated largely from open wells. This is explained by a second effect which relates to a phenomenon highlighted by Prasad and Sohoni, 2020. Farmers invest in irrigation setups such as borewells and long distance

Preprint of: https://doi.org/10.1016/j.esd.2025.101719

pumping from surface water sources, to assure water for high value crops. High value crops include perennial and rabi horticultural crops, and not rabi Sorghum and Gram which the researchers say are the lowest receiving priority. These two crops may also be cultivated by farmers who do not have access to irrigation (Bhopale, 2019; Nimbalkar et al., 2016;).

The survey data showing correlation of crops to water sources is given in table A.10. The crops have been categorized based on their season into (i) Kharif and long kharif crops, (ii) Gram and rabi Sorghum, (iii) Other rabi crops, and the yearlong (iv) Sugarcane and Plantation crops. The area covered by crops in the state, listed in table A.10, shows the predominance of the first two categories, which is broadly in keeping with the large proportion of open wells compared to assured water sources.

Hence in the methodology, for kharif and long kharif crops, Gram, and rabi Sorghum, energy estimation is done on the basis of open well energy intensity. For the other crops, the trend of open wells drying out among farmers doing rabi cultivation is considered to estimate the quantity of irrigation water being sourced from assured water sources. For example in cluster 1, 60% open wells are dry in November, increasing to 100% in April. Hence, in November 60% of the irrigation water is estimated to be from borewells, and the rest from open wells with this percentage increasing over the season in a linear trend. Borewell energy intensity is used for that percentage of water when estimating the energy used. Cluster 4 and 5 are examples where the assured water sources are a canal and river respectively, instead of borewells. The crop - water source correlations observed in the 5 clusters are listed in table A.11.

5.5 Water transfers

Water is not pumped into the fields directly from borewells because the flow rate is too low. Instead the water is first transferred into open wells and then used for irrigation. This was observed for all farmers using borewells. Such transfers have also been observed in Gujarat (Kumar et al. ,2022).

In cluster 4 farmers have an additional reason for transfers. Canal releases in the area take place once a month, however horticultural crops irrigated by drip need to be irrigated every few days. Farmers pump water from the canal during a release into their storage tanks and open wells which are close to their fields and use as required every day. These transfers need to be considered for energy intensity calculations.

The water transfers observed in the 5 clusters are listed in table A.11.

5.6 Determining the crop-wise water and energy usage

The water and energy usage parameters developed in the preceding sections are applied to each crop and irrigation method combination in each cluster to determine the water and energy usage per unit area for a crop. As an example, this calculation has been described here, for wheat in

cluster 1. In this paper water usage refers to the irrigation water applied to the field and not actual evapotranspiration.

Wheat is irrigated by the border method. The number of irrigation events for wheat in cluster 1 from table A.5, 9.1, and irrigation depth for the border method from table 1, 54 mm (and 66 mm for the first irrigation), are multiplied to find the water applied. This is found to be 510 mm. The irrigation events are assumed to be uniformly distributed over the crop season from November to March (season is provided in table A.5), and hence water usage turns out to be 102 mm per month.

This 102 mm is assigned to one or more water sources according to the crop-water source correlation and the seasonality of the water sources. In cluster 1, open wells and borewells are used for wheat. The proportion of borewell water increases from 40% in November to 88% in March (from section 5.4.1). The energy intensity of borewells (section 5.1.2) and the volume of water from borewells, is used to find the energy used in a month. For example, in November, 102 mm x 40%, or 41mm, is sourced from borewells, and the energy intensity in November is 0.5 kWh/m^3. Hence the energy for borewell pumping is 205 kWh/ha.

Since water is transferred from borewell to open well and then applied to the field, all of the water needs to be pumped from the open well. Hence 102 mm of water, and an energy intensity of unpressurised pumping 0.26 kWh/m^3 (from section 5.1.1), are multiplied, resulting in an energy usage of 265 kWh/ha in each month for open wells. The energy consumption from both sources, from November to March, are totalled to find the energy consumption over a season.

This methodology is applied to all crops in the five clusters to estimate the water and energy usage. The outputs for clusters 1 to 5 are listed in Appendix A in tables A.12 to A.16 respectively. The outputs of the methodology include, for each crop-irrigation method combination in a cluster: water usage in mm, energy usage in kWh/ha, and energy intensity of irrigation in kWh/m^3.

6. Results: Energy and water usage in the study area

Figure 7 illustrates the energy intensity of irrigation in kWh/m^3, irrigation water applied in mm, and energy usage per unit area over the season in kWh/ha, in box plots, aggregated under the four crop categories developed in section 5.4. Some subcategories have been created based on the distinctions visible in the data for each output type. The data points comprising the box plots include individual farmer measurements, and the methodology average outputs for the 5 clusters. Energy intensity data comprises average energy intensity over a season. If a crop is irrigated by more than one water source, the energy intensity indicates the weighted average over all sources.

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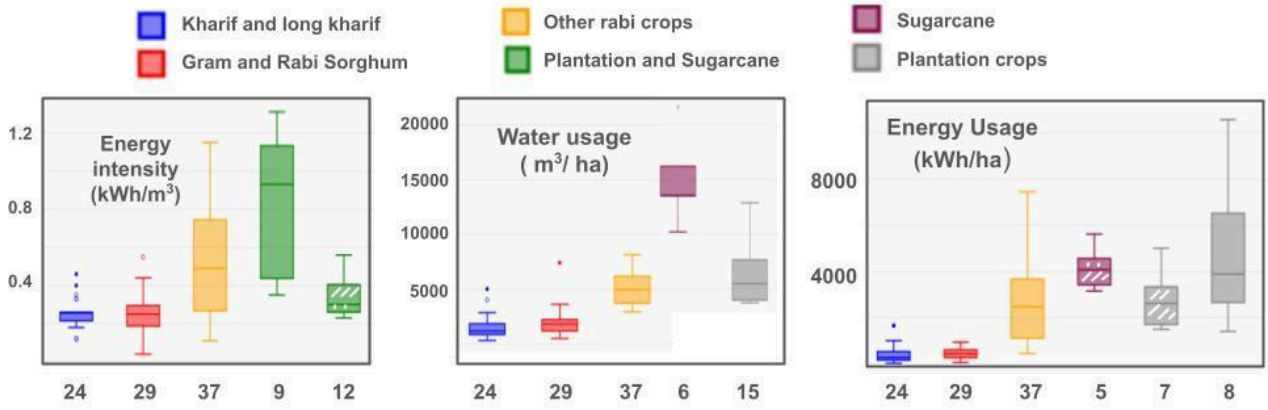


Figure 7. Measured and estimated energy and water usage data. The number of data points in the box plots are indicated below each plot. The box plots which have a white hatched pattern indicate irrigation from surface water sources while the rest are groundwater irrigated.

In the energy intensity plot, 'Plantation and Sugarcane' has been separated into groundwater and surface water source irrigation data. Surface water irrigation means the crop may be irrigated only by surface water sources, or by open wells in kharif and surface water when the open wells dry up as explained in section 5.4.

'Kharif and long kharif' crops, and 'Gram and rabi Sorghum' exhibit very similar ranges of energy intensities between 0.1 to 0.5 kWh/m^3 and averages of 0.25 and 0.26 kWh/m^3 respectively.

The energy intensity for 'Other rabi crops' ranges from 0.1 to 1.2 kWh/m^3 with an average of 0.53 kWh/m^3 . For 'Plantation and Sugarcane' crops, for groundwater sources it ranges from 0.3 to 1.3 with an average of 0.82 kWh/m^3, and for surface water sources it ranges from ~0.2 to 0.6 kWh/m^3 with an average of 0.34 kWh/m^3.

This breakup of groundwater and surface water sources has not been done for the first two categories because they are largely irrigated from open wells. For the 'Other rabi crops' category, the surface water and groundwater irrigated data has not been separated out since only 5 out of 37 data points are from surface water sources, and the overall distribution doesn't change much even if the 5 surface water data points are removed. The surface water energy intensity for 'Other rabi crops' is similar to that observed for surface water sources with 'Plantation and Sugarcane'.

The average water usage for 'Kharif and long kharif' and 'Gram and rabi Sorghum' do not indicate a distinction. As expected, it is higher for 'Other rabi crops'. The water usage for 'Plantation and Sugarcane' has been split into Plantation crops, and Sugarcane, since the two subcategories show a clear distinction. Plantation crops have higher values of water usage than 'Other rabi crops'. Sugarcane is clearly much higher than all other crops at 10,000 to 17,000

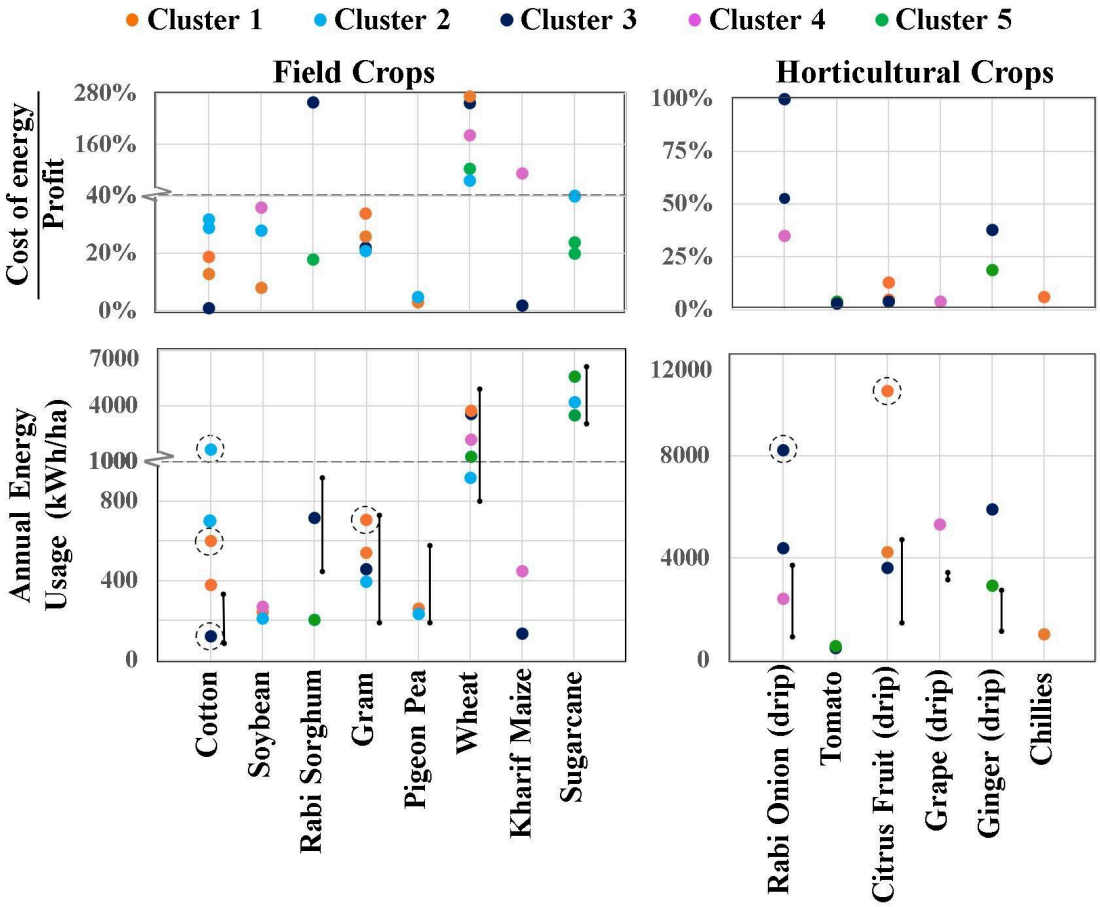
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m^3/ha. There is no distinction visible in the water usage (m^3/ha) by groundwater and surface water sources for all categories, and hence are not disaggregated in the plot.

The energy usage for the 'Kharif and long kharif', and 'Gram and rabi Sorghum' categories is less than 1000 kWh/ha, with averages of 400 and 440 kWh/ha respectively. 'Other rabi crops' exhibit a wide range with an average of 2604 kWh/ha, and the averages for groundwater and surface water irrigation separately, not illustrated in the plot, are 2680 kWh/ha and 2120 kWh/ha respectively. As in the case of the energy intensity plot, surface water irrigated here implies that a data point could include open well irrigation in addition to surface water.

For sugarcane only the surface water sourced data points are included in the box plot because 5 out of the 6 data points are based on surface water sources. Average sugarcane energy usage is 4115 kWh/ha. For plantation crops the average energy usage for surface water and groundwater source irrigation are 2696 and 4763 kWh/ha respectively.

Figure 8 illustrates the crop-wise energy usage over the season in kWh/ha and cost of energy as a percentage of the profits. Each data point is the estimated average energy usage in a cluster for a particular crop due to variations in the number of irrigations, irrigation methods, and water sources.



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Figure 8. Crop-wise energy usage and cost of energy as a % of profit as a function of the cluster. The horizontal dashed lines in the field crops plots mark a change in scale. The black vertical lines in the energy usage plots show the range of measured energy consumption of the metered farmers. An irrigation method has been mentioned with the crop name in case of pressurized irrigation. The dashed circles indicate data points for which the irrigation method is different from the method indicated with the crop name. These irrigation methods are: drip for Cotton, sprinkler for Gram, border for Rabi Onion, and furrow for Citrus Fruit.

Profits have been determined from surveys for clusters 1 and 3. For clusters 2,4 and 5, profits have been calculated from state level data (CACP, 2018; CACP, 2017). All operational costs including imputed family labor have been considered. Energy costs are not included since most farmers do not pay bills. Profits are listed in tables A.12 to A.16 for the respective clusters.

7. Discussion

A methodology has been developed quantifying the WEF interlinkages such that it can be applied to a location by collecting some data through surveys. We discuss how the methodology can be applied to a location.

7.1 Applying the methodology

Crop-wise energy and water usage for irrigation was estimated for 5 clusters. The same methodology may be applied to any area of the state. The parameters categorized under local irrigation practices need to be determined from surveys, i.e. the crop-wise number of irrigations and irrigation methods, the seasonality and availability of water sources. This data, along with the energy intensities and the irrigation depth parameters developed in section 5, can be used to find the crop, energy, and water linkage. This is illustrated in an example below.

21% of the sugarcane in the state is cultivated in the Marathwada division, with irrigation from open wells and borewells. Subsequent work conducted in the PoCRA project has shown that furrow irrigation is the commonly used method for sugarcane in Marathwada with 24 to 30 irrigation events in a year. Open wells run out of water between November and February, after which borewells are used. The irrigation depth for furrow method, and energy intensities for groundwater sources developed in sections 5.1 and 5.2, are used to estimate annual water usage at 1080 to 1350mm, and energy usage between 7470 and 12,308 kWh/ha. This is two to three times the energy usage for sugarcane cultivated from surface water sources in Western Maharashtra as illustrated by the example of cluster 5 at 3510 and 5616 kWh/ha listed in table A.16.

Since the methodology is based on the number of irrigations, the rainfall in a particular year or other weather conditions do not affect the application of the methodology. Though the outcomes, i.e., the energy and water usage, will be affected.

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7.2 Output of the methodology: Energy and water usage by crops

The results obtained through application of the methodology developed in this work quantify various energy usage parameters that have largely not been quantified, or documented heretofore. This is an important contribution.

Fig.8 illustrates the crop-wise energy usage estimates from the methodology in kWh/ha as function of cluster along with the range of measured energy consumption where available. The cluster estimates are broadly seen to be a good match with the measurements.

Fig.7 shows the broad trends and ranges of energy intensity, and energy and water usage by crop category. We discuss the main results of these trends in this section.

'Kharif and long kharif' crops, and 'Gram and rabi Sorghum' exhibit average energy intensities of ~0.25 kWh/m^3, which is close to the average energy intensity of unpressurized pumping from open wells. It needs to be emphasized here that a large proportion of kharif farmers do rainfed farming and hence are not a part of the sample of this study. This is reflected in the number of pump connections which are equal to ⅓ the number of landholdings in the state, while kharif and long kharif crops cover 65% of the net cultivable area of the state.

Field measurements illustrated in fig.4 show an average energy intensity of pressurized irrigation from open wells being 21% higher than that of unpressurized irrigation. But this distinction is not visible for the energy intensity of the first two categories which are largely irrigated from open wells, possibly because of insufficient data points.

Surface water sources for plantation and sugarcane, exhibit an average of 0.34 kWh/m^3. This is much lower than the groundwater average of 0.82 kWh/m^3 which is high due to a substantial contribution from borewells. For plantation crops, surface water energy consumption is ~60% of groundwater energy consumption. For sugarcane, surface water energy consumption is 35% to 55% of the groundwater energy consumption, estimated for Marathawda in section 7.1. This is relevant since farmers tend to cultivate such water intensive crops if they have access to assured water from surface water sources or borewells. And this result highlights the difference in energy consumption between the two types of sources.

7.3 Application to tariff policy

Researchers have proposed that removal of subsidies will lead to reduced irrigation water and energy usage (Shah, Giordano, and Mukherji 2012; Badiani and Jessoe, 2011; Pachauri and Mehrotra, 2002). We analyze the potential effect of removal of subsidies considering the cost of energy as a percentage of profit illustrated in fig.8.

Among the field crops, Wheat will become uneconomical when the cost of energy is factored in, since cost of energy is higher than profits in most locations. Among horticultural crops, the

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cost of energy for drip irrigated rabi Onion at 50% of current profit is quite high and for the border irrigated crop it is 100%. Subsidy removal will render rabi onion uneconomical, or at best too risky when the high degree of fluctuation in the market price for horticultural crops is factored in (Prasad and Sohoni, 2020). On the other hand, the plantation crops, Grape and Orange, will not be affected by subsidy removal since the profits are high and the cost of energy is less than 10%. These two effects together may lead to a type of rebound effect as follows.

As mentioned in section 5.4, the water resource in the state is replenished every monsoon with a limited availability (Barik et al., 2017). The reduction in water usage by wheat and crops like rabi Onion, may lead to an increase in the availability of groundwater for a longer period after the monsoons. It has been observed that availability of water often limits the area under cultivation of horticultural crops and Sugarcane, hence a greater availability of water may lead to an increase in cultivation of these crops. In such an event subsidy removal will not lead to a reduction in energy or water usage but only a redistribution. This potential effect of a move to high valued crops as a result of increased tariff has been noted by Kumar et al. (2007).

The cost of energy for irrigation estimated through this methodology provides inputs for a more realistic economic analysis of the effect of energy pricing on food security and farmer incomes. So far, simplified data has been used for such exercises (Bhattacharyya, and Ganguly, 2017; Swain and Charnoz, 2012; Sant and Dixit, 1996). For example, Bhattacharya and Ganguly (2017), estimate the effect of removal of power subsidy on food prices and inflation using data from the national Social Accounting Matrix, where agricultural output is connected to electricity input based on country wide electricity usage in agriculture, hence using one average electricity input figure for all crops in the country. As seen from the wide variations in the energy input for different crops in this work, the approach followed by Bhattacharya and Ganguly (2017) will result in unrealistic outcomes.

7.4 Effect of pressurized irrigation on energy intensity

Average energy intensities for pressurized and unpressurized irrigation from open wells are found to be 0.26 kWh/m^3 and 0.33 kWh/m^3. The effect of increase in energy usage due to pressurized irrigation has never been quantified in the Indian context. The effect of the Best Efficiency Point (BEP) and the very low efficiency operation of unpressurized pumping at 5 to 10%, shows that using an average efficiency number as done in previous studies would lead to erroneous results in energy usage estimations.

This also highlights the importance of updating agricultural pump standards. Currently the standards are designed for a single BEP. With the common usage of sprinklers, it is clear that farmers commonly operate their pumps at two very different operating points, pressurized irrigation with sprinklers, and unpressurized irrigation methods such as furrow and border. This practice calls for an updated approach for efficiency and labeling standards.

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The quantification of irrigation depths or water applied, and the energy intensities, together clearly bring to light energy-water trade-offs in using pressurized irrigation. For instance, sprinkler irrigation which is commonly used for gram results in 11% water usage reduction from the border method, but an increase in energy usage of 13%. Whereas in the case of orange, a move to drip from furrow irrigation reduces both water as well as energy usage by 2.5 times. Schemes that expect to encourage water savings technologies through energy payback, need to take such aspects into consideration.

7.5 Further development of the methodology

This work develops a methodology to determine the WEF interlinkages, incorporating local irrigation practices. Dataset development of these practices by village, such as the number of irrigations and irrigation methods, and crop - water source correlation, can aid policy design and evaluation of the effect of policies over time. For instance, the current government policy of promoting horticulture may result in more farmers extracting water after the kharif season thus leading to a need for borewells, and nudging a location like cluster 2 with only open wells, closer to a situation like clusters 1 and 3 with borewells. These trends may be identified through such datasets and can be used to fine-tune policies. Nexus researchers have called for the development of such site-specific fine-grained data (Zhang et al., 2018b; Stillwell et al., 2011).

The energy intensity for open well and borewell has been developed as averages for energy intensity because of the commonality in well depths and pump HPs across the state. In contrast, the energy intensity for surface water sources needs site specific information such as pipe diameters, and elevation changes. Further investigations to standardize the surface water energy intensities as a function of easily determined factors such as typical flow rates and pump HPs, combined with empirical data, can further simplify the methodology in case of surface water sources.

The common irrigation depth categories have been developed in this work, but further categorizations may also be required depending on the application. For instance, if the irrigation water reduction potential of drip irrigation for rabi onion is to be investigated, a larger dataset specific to the crop is needed.

8. Conclusion

A detailed methodology incorporating hydrogeology and local irrigation practices has been developed for small-holder farmers with private irrigation systems. The results include the energy, water, and crop linkages for the main crops in Maharashtra, India. Field level energy and water usage measurements support the development of the methodology and the outcomes.

The energy intensity (kWh/m^3) for water sources in the state has been quantified. Herefore, there has been very little published data on energy usage at the farmer level. Two main local irrigation practices that affect the WEF linkages, namely the number of irrigation events, and correlation

of crops to water sources, have been identified and formalized through 5 representative locations in the state.

The paper points out the exceptional values and divergence of energy intensity observed for some crops and the practices behind these. Some policies being proposed to reduce energy and water usage in irrigation have been analyzed using the results of the methodology.

Farmers are doing pressurized and unpressurized irrigation for different crops with a pump suitable for the high head required in pressurized irrigation. Thus resulting in very low energy efficiency operation for unpressurized irrigation.

The study reveals the complex irrigation practices and factors that determine energy usage in irrigation for a crop, hence highlighting the importance of conducting research and developing datasets, at a local level in the design of policies.

The generalized nature of the methodology ensures that it can be applied across the state and is not limited to site-specific cases. It can also be extended to other regions with similar small-holder irrigation, through the adaptation of the physical parameters.

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